

Course Outline

AWS Academy Generative AI Foundations

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Course version

This course outline applies to version 1.0 of AWS Academy Generative AI Foundations course in English.

Description

AWS Academy Generative AI Foundations course offers an introduction to artificial intelligence with a particular emphasis on generative AI technologies. While providing a broad foundational understanding of AI and Machine Learning (ML), the curriculum delves deeper into the concepts of generative AI and its applications in cloud computing.

Students will explore the fundamental concepts, capabilities, principles, use cases and associated AWS services and tools of generative AI in the context of the AWS cloud.

The concepts in this course are aligned to the generative AI concepts covered in AWS Certified AI Practitioner (AIF-C01) exam but is not designed as preparation for this exam.

Curriculum objectives

This course prepares students to do the following:

- Summarize generative AI concepts, methods, and strategies.
- Discuss the potential business value and use cases of generative AI.
- Match AWS AI/ML services and features to generative AI use cases.
- Explain how to use foundation models.
- Discuss techniques to refine results of foundation models.
- Recognize the key elements for practicing responsible AI.
- Discuss the security, privacy, and governance considerations for AI systems.
- Describe the workflow and key considerations when developing a generative AI application.

Duration

The duration of the course is approximately 12-15 hours when delivered synchronously by an educator. This course is designed to be delivered over one semester. Actual delivery times will vary from class to class and depend on delivery format.

Intended audience

This foundational (level 100) course is intended for students at AWS Academy member institutions who seek an overall introduction to generative AI concepts, strategies, and the associated AWS services and tools.

Student prerequisites

To ensure success in this course, students should have knowledge of the concepts covered in the AWS Academy Cloud Foundations course or have equivalent experience.

Delivery methods

Learning materials are provided to support synchronous or asynchronous learning. Lecture slides and an instructor guide are provided for instructor-led training. Recorded lectures and demos are provided for independent learning. The educator can determine the preferred delivery method for each module.

Educator prerequisites

This course does not have any prerequisites for educators. However, prior to facilitating this course, educators are recommended to complete this course and complete the AWS Academy Cloud Foundations course.

Learning resources

- Lecture slides
- Student guide
- Instructor guide
- Instructor-led activities
- Recorded lectures
- Recorded demos
- Module knowledge checks
- Course assessment

Course timing

This table provides the suggested durations for all course activities. The total classroom time for all the modules in this course is 900 minutes. Items that are not applicable are marked NA.

Module Title	Lecture (Minutes)	Activity, Lab, Demo, Sample Question (Minutes)	Knowledge Check (Minutes)	Total Module (Minutes)
Module 1: Welcome to AWS Academy Generative AI Foundations	30	NA	NA	30
Module 2: Introducing AI	45	10	10	65
Module 3: Introducing Generative AI	90	20	10	120
Module 4: Using Prompts and Prompt Engineering	100	35	10	145
Module 5: Working with Foundation Models	120	25	10	155
Module 6: Practicing Generative AI Responsibly	90	15	10	115
Module 7: Considering Security, Governance, and Compliance for Generative AI	90	15	10	115
Module 8: Implementing Generative AI	100	5	10	115
Module 9: Bridging to Certification	10	NA	NA	10
Course Assessment	NA	30	NA	30

Module Title	Lecture (Minutes)	Activity, Lab, Demo, Sample Question (Minutes)	Knowledge Check (Minutes)	Total Module (Minutes)
Total Course Time	675	155	70	900

Module learning objectives and sections

This section lists the learning objectives and sections in the modules of this course.

Module 1: Welcome to AWS Academy Generative AI Foundations

Module sections

- About this course
- Course outline

Module 2: Introducing AI

Learning objectives

- Define AI and machine learning (ML) terminology.
- Explain the relationship between the concepts of AI, ML, deep learning, and generative AI.
- Distinguish between scenarios that are suited for traditional software methods, traditional ML methods, or generative AI.
- Identify the capabilities of AWS AI/ML services.

Module sections

- AI overview
- Machine learning concepts
- Machine learning, deep learning, and generative AI
- AWS AI services
- Demo: Using AWS AI Services
- Knowledge check

Module 3: Introducing Generative AI

Learning objectives

- Describe key characteristics of foundation models (FMs).
- Define a prompt in the context of generative AI.
- Describe task types for large language models (LLMs).
- Explain the role of tokens, embeddings, and vectors in LLMs.
- Recognize the potential business value of generative AI.
- Summarize key challenges of using generative AI, and discuss how to mitigate them.
- Identify methods for using additional data to influence the outputs of an FM.
- Describe three primary AWS services for generative AI.
- Identify potential use cases for generative AI.

Module sections

- Foundation models
- How foundation models work
- AWS generative AI services
- Demo: Using Amazon Q Developer
- Generative AI use cases
- PartyRock, an Amazon Bedrock playground

- Activity: Introducing PartyRock
- Knowledge check

Module 4: Using Prompts and Prompt Engineering

Learning objectives

- Identify best practices for designing effective prompts.
- Define prompt engineering.
- Define common elements of a prompt.
- Describe fundamental prompt engineering techniques.
- Identify potential prompt misuses.
- List key Amazon Bedrock features related to prompt engineering.

Module sections

- The value of prompt engineering
- Structuring a large language model prompt
- Activity: Structuring an effective prompt
- Prompting techniques
- Activity: Applying prompting techniques
- The risk of adverse prompts
- Amazon Bedrock features for prompt engineering
- Knowledge check

Module 5: Working with Foundation Models

Learning objectives

- Discuss the factors to consider when selecting and evaluating a foundation model (FM).
- List common inference parameters, and describe their effect on model outputs.
- Describe Retrieval-Augmented Generation (RAG).
- Distinguish between FM customization techniques, including fine-tuning and continuous pre-training.
- Recognize key features of Amazon Bedrock and Amazon SageMaker AI for working with FMs.

Module sections

- Considerations for choosing and adapting a foundation model
- Choosing the right foundation model for a use case
- Evaluation and monitoring
- Configuring inference parameters
- Activity: Adjusting inference parameters
- Retrieval-Augmented Generation (RAG)
- Foundation model customization techniques
- Working with foundation models on AWS
- Demo: Working with Models in Amazon Bedrock
- Knowledge check

Module 6: Practicing Generative AI Responsibly

Learning objectives

- Define the core dimensions of responsible AI.
- Discuss the challenges of generative AI.
- Discuss the trade-offs of transparency, interpretability, and explainability.
- Identify AWS services and tools that support building responsible AI solutions.

Module sections

- Defining responsible AI
- Generative AI considerations
- Implementing safeguards
- How AWS generative AI services support responsible AI
- Demo: Using Amazon Bedrock Guardrails
- Knowledge check

Module 7: Considering Security, Governance, and Compliance for Generative AI

Learning objectives

- Discuss the security, privacy, and governance considerations for AI systems.
- Describe key aspects of a defense in depth approach and how AWS supports this approach.
- Identify AWS services that assist with applying governance controls and achieving compliance objectives in AI solutions.

Module sections

- Defining security, governance, and compliance for generative AI
- Security of AI systems on AWS
- Governance of AI systems on AWS
- Compliance of AI systems on AWS
- Demo: Using AWS Audit Manager Frameworks
- Knowledge check

Module 8: Implementing Generative AI Applications

Learning objectives

- Discuss the workflow to develop a generative AI application.
- Review key considerations when selecting a foundation model (FM).
- List metrics to evaluate performance of foundation models.
- Map AWS generative AI services and a generative AI workflow for a given use case.

Module sections

- Generative AI application lifecycle
- Using AI agents in Amazon Bedrock
- Matching the AWS service to the use case
- Knowledge check

Module 9: Bridging to Certification

Learning objectives

- Identify how to prepare for the AWS Certified AI Practitioner (AIF-C01) exam
- Find resources to prepare for the exam

Module sections

- AWS Certified AI Practitioner (AIF-C01) – Exam details
- AWS Certified AI Practitioner (AIF-C01) – Exam resources